

Nonradiative carrier capture by point defects in irradiated GaN

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ABSTRACT

In this work, accurate electronic structure calculations are employed to investigate key radiation-relevant defects in gallium nitride, including point defects (vacancies, interstitials, and antisites) as well as the gallium–nitrogen divacancy. Nonradiative carrier capture via multiphonon emission was explored for each of the thermodynamic charge transition levels identified for these defects. Benchmarking was performed using a carbon substitutional defect on a nitrogen site, yielding results that agree well with previous electronic structure and photoluminescence results. Seven defect charge transitions were found to have capture coefficients above 10^{-10} cm³/s at 300 K: octahedral Ga_i (+3/+2) electron capture, octahedral Ga_i (+2/+1) electron capture, tetrahedral Ga_i (+1/0) electron capture, V_N (+1/0) electron capture, V_{Ga} (0/+1) hole capture, split N_i (0/−1) electron capture, and V_N (0/−1) electron capture. These transitions play a central role in governing defect charge-state populations and the resulting carrier recombination dynamics under irradiation. Previous displacement damage studies found relatively high concentrations of vacancies and nitrogen interstitials; therefore, V_{Ga} (0/+1) hole capture, split N_i (0/−1) electron capture, and V_N (0/−1) electron capture are likely key contributors to radiation-induced performance degradation in GaN devices.

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I. INTRODUCTION

Gallium nitride (GaN)-based electronic devices have been considered radiation-hard alternatives to Si-based devices for space applications.¹ As a wide-bandgap semiconductor material, defects produced in GaN, such as thermally or by radiation, can act as traps for charge carriers. Dependent on the local chemical environment and Fermi level, defects have energetically favorable charge states that they transition toward through capture and emission of charge carriers (electrons and holes). Carrier capture processes may be either radiative or nonradiative during Shockley–Read–Hall (SRH) recombination.^{2–4} Nonradiative capture can dissipate energy through either the Auger effect, multiphonon emission (MPE), or cascade capture.⁵

Carrier capture occurs because the energy level of the defect varies based on the local environment. Capture into the new defect charge state changes the energy profile with respect to the original local structure. After capture occurs, the defect attempts

to relax into this new equilibrium configuration to minimize the system energy. For MPE, this relaxation occurs via emission of lattice phonons to dissipate the excess energy. A common phenomenon in semiconductors including III–V compounds, MPE can lead to large capture cross sections as well as low-temperature defect motion. Such removal of carriers by charge trapping causes a reduction in carrier density and carrier mobility. This can negatively impact device performance due to shifts in the threshold voltage, decreases in ON-state current, or decreases in transconductance.⁶ Therefore, understanding charge trapping due to MPE by radiation-induced defects is crucial for developing GaN-based and Al_xGa_{1−x}N devices.¹

Shi and Wang⁷ presented an *ab initio* approach for determining nonradiative recombination rates in bulk semiconductors and applied it to hole capture for the (0/+1) transition by Zn_{Ga}–V_N in GaN. The approach incorporated electronic structure calculations for both electron transitions and electron–phonon coupling matrices. This yielded results that aligned with the low end of

experimentally fit nonradiative capture rates for a blue shift in photoluminescence (PL).^{8–12} Shi and Wang's methodology was expanded by Alkauskas *et al.*¹³ using hybrid density-functional calculations and a static-coupling approach,^{14,15} as opposed to the adiabatic approach,¹⁶ alongside the Condon approximation. Details on this approach are included in Sec. S B of the [supplementary material](#). Using this technique, Alkauskas *et al.* computed hole capture coefficients for the (0/+1) transition of the $\text{Zn}_{\text{Ga}}\text{-V}_{\text{N}}$ complex and the (−1/0) transition of C_{N} in GaN. For $\text{Zn}_{\text{Ga}}\text{-V}_{\text{N}}$, Alkauskas *et al.* found hole capture coefficients roughly an order of magnitude or two greater across the temperature range as compared to those from Shi and Wang, with the room-temperature value near the middle of the experimentally identified range.^{8,11} Previous studies indicate that C_{N} is responsible for the yellow photoluminescence (PL) band in GaN.^{17–19} The hole capture coefficients by C_{N} were determined between 500 and 700 K from time-resolved PL measurements.^{20,21} There was approximate agreement between these experimentally determined values and those calculated by Alkauskas *et al.*¹³ when several thermal effects are taken into account.

More recently, Zhao *et al.*²² determined nonradiative capture coefficients, radiative capture coefficients, and trap-assisted Auger-Meitner (TAAM) coefficients for the (0/+1) and (−1/0) transitions of C_{N} as well as the hole thermal emission coefficient for (+1/0). For nonradiative capture coefficients, the same approach as Alkauskas *et al.* was employed, while additional methods were used for the TAAM²³ and trap-assisted radiative²⁴ coefficients. The nonradiative hole capture processes for both (0/+1) and (−1/0) are dominant compared to the radiative capture, while the inverse is true for the electron capture processes. The total trap-assisted radiative recombination rate was then determined including MPE, radiative, TAAM, and thermal emission across both transition levels to demonstrate the overall impact of C_{N} defects in GaN-based devices. Nichols *et al.*²⁵ also developed a first-principles methodology for nonradiative carrier capture calculations based on the theory of inelastic multiphonon carrier capture process to determine zeroth- and first-order contributions within the adiabatic, Born-Oppenheimer approximation, similar to the work of Barmparis *et al.*²⁶ Example calculations were presented for (−1/0) hole capture by C_{N} in GaN, which demonstrated good agreement with experimental results around 600 K.^{10,12,27,28}

During irradiation, all kinds of point defects may exist in GaN. Therefore, there is a need to have a consistent set of results for nonradiative recombination rates to understand the relative process likelihoods and impacts. To accomplish this, the main theoretical framework from Alkauskas *et al.* was followed, with three major differences. Firstly, anharmonic spline fits were generated for potential energy curves rather than assuming the harmonic approximation. Secondly, electron-phonon coupling matrix elements were calculated within the projector-augmented-wave (PAW) formalism instead of switching to the plane-wave pseudo-potential formalism. Finally, a more robust scaling parameter was used, more accurately reflecting charged defect considerations both for temperature and computational artifacts. Alignment of the methodology used here with previous work was demonstrated using the C_{N} defect, discussed below. For calculating carrier capture rates for important point defects in GaN, MPE is solely

considered since, for many cases, MPE capture rates can be much larger than radiative rates.¹³ Six electron capture processes [octahedral Ga_i (+3/+2), octahedral Ga_i (+2/+1), tetrahedral Ga_i (+1/0), V_{N} (+1/0), split N_i (0/−1), and V_{N} (0/−1)] and one hole capture process [V_{Ga} (0/+1)] were identified as particularly significant at 300 K. Note, however, that some other capture coefficients were found within or below the expected range of the radiative capture coefficients.²³ Carrier capture coefficients determined in this work can be applied directly to understand how the capture processes for various defects impact carrier concentrations in devices.^{29–31}

II. METHODS

Electronic structure calculations were performed using density-functional theory (DFT) as implemented in VASP.^{32–34} To reduce computational costs, core electrons were treated with pseudopotentials using the PAW formalism.^{35,36} For Ga, an explicit valence electronic configuration of $4s^2 4p^1$ was used, leaving the $3d$ semicore states in the core. A plane-wave basis set cutoff energy (ENCUT) of 400 eV was used for all calculations. To properly compute electronic properties, specifically the electronic bandgap, the hybrid functional proposed by Heyd-Scuseria-Ernzerhof (HSE06)^{37,38} was used to more accurately account for the exchange-correlation effects. The HSE fraction of exact exchange and the screening parameter were $\alpha = 0.305$ and $\mu = 0.2 \text{ \AA}^{-1}$, respectively.

For modeling point defects, a 96-atom ($3 \times 2 \times 2$) supercell of orthorhombic gallium nitride (GaN) was constructed. Orthorhombic unit cell lattice parameters optimized at the HSE-level via equation of state fitting with a $4 \times 4 \times 4$ k-point mesh were found to be $a = 3.195 \text{ \AA}$, $b = 5.534 \text{ \AA}$, and $c = 5.189 \text{ \AA}$. These lattice parameters and electronic bandgap, 3.518 eV, show good agreement with previous experimental results.³⁹ Nearly all the same calculation parameters as used by Alkauskas *et al.* were used in this work and are summarized below.¹³

Defect relaxation was performed to determine equilibrium structures for each defect charge state. A single special k-point ($\frac{1}{2}, \frac{1}{2}, \frac{1}{2}$) was chosen for its comparable accuracy with a $2 \times 2 \times 2$ k-point mesh through testing. Both the Ga valence configuration and the special k-point were chosen to improve computational efficiency while maintaining accuracy. Associated defect system energies were used in Eq. (S1) of the [supplementary material](#) to construct formation energy (E_f) diagrams as a function of Fermi level (E_F). Chemical potential bounds were determined as described in Eq. (S2) of the [supplementary material](#), where the reference calculations (details provided in Sec. S A 1 of the [supplementary material](#)) used tools from both the Materials Project^{40–43} and pydefect.⁴⁴ Ga-rich conditions were chosen for all charge transition values reported here. Defect analysis was performed with pydefect⁴⁴ and doped,⁴⁵ utilizing the extended Freysoldt-Neugebauer-Van de Walle (eFNV) scheme devised by Kumagai and Oba⁴⁶ to account for finite-size charge corrections.

Configuration coordinate diagrams (CCDs) represent the 1D potential energy profile of each defect charge state within a pair involved in a transition. A mass-weighted configuration coordinate, Q , was first defined based on the difference between equilibrium configurations, ΔQ . Q was then used to interpolate and extrapolate

around the two equilibrium configurations, executing single-point energy calculations for variable Q configurations. `CarrierCapture.jl`⁴⁷ fit spline potentials to the displaced calculation data to construct the potential energy curves. The final state curve was shifted to have the equilibrium energy equal to the valence band maximum (VBM). A second final state curve was offset by the bandgap (E_g) to represent the energy of a hole at the VBM and an electron at the conduction band minimum (CBM). The initial state curve was offset by the ionization energy, ΔE , modeling the energy of one of the carriers. ΔE is equal to the charge transition level (CTL) relative to the VBM, taken from formation energy diagrams.⁴⁸ Classical capture barriers (ΔE_b) are the energy of the crossing point between the two potential energy curves involved in the capture process, relative to the equilibrium energy of the upper curve.⁴⁹ The values are, therefore, dictated by the potential energy curve fits and ΔE .

Following this, electron–phonon coupling matrix elements were computed using a two-step process, still within the PAW formalism.⁵⁰ The one-dimensional approximation simplifies consideration of all phonon modes to only the strongest accepting phonon mode (strongest coupling to the configurational change).¹³ MPE can then be modeled through a single electron–phonon coupling matrix element between the charge carrier state at the valence/conduction band edge and the defect state in the gap involved in the capture process. Since GaN has a direct bandgap at the Γ -point, the wavefunctions at this point are needed. Following Alkauskas *et al.*,¹³ Ga $3d$ states were included in the valence configuration ($4s^2 3d^{10} 4p^1$) for electron–phonon coupling matrix calculations. The supercell bandgap was found to be 3.513 eV in this case. Single-point energy calculations using a Γ -point only k -point mesh were performed for several configuration coordinate points around the chosen reference structure (typically the neutrally charged or less charged equilibrium structure). Here, two configurations on either side of the reference were selected with $\delta Q = 0.1 \text{ amu}^{1/2} \text{ \AA}$ between each. $\tilde{W}_{if}$ values were calculated in a second step using the process detailed in `nonrad`.⁵⁰ Seen in Eq. (S5) of the [supplementary material](#), the electron–phonon coupling matrix was constructed based on the overlap between the reference configuration wavefunction and each wavefunction from the set of five configurations including and around the reference. VASP postprocessing (ALGO = None) generated the matrix (LWSWQ = True) between the WAVECARs from the previous step and the reference. For GaN, the valence band is (nearly) triply degenerate, and therefore, the root mean square was taken for the three corresponding values when considering hole capture. Using C_N ($-1/0$) as an example, the $\tilde{W}_{if}$ values for hole capture and electron capture were found to be 9.13×10^{-2} and $4.63 \times 10^{-3} \text{ eV/amu}^{1/2} \text{ \AA}$, respectively, with the $3d$ states of Ga in the valence configuration. These values increase by less than 10% to 9.29×10^{-2} and $5.07 \times 10^{-3} \text{ eV/amu}^{1/2} \text{ \AA}$ when leaving the $3d$ states in the core.

Carrier capture coefficients [Eq. (S3) in the [supplementary material](#)] were calculated from 10 to 800 K using the `CarrierCapture.jl`⁴⁷ and `nonrad`⁵⁰ codes. `CarrierCapture.jl` solved the 1D Schrödinger equation for the quantum oscillators, evaluated the overlap between phonon wavefunctions, and subsequently calculated the capture

coefficients. `nonrad` determined the electron–phonon coupling matrix elements and scaling factors. If the reference configuration for electron–phonon coupling was associated with the neutral state, the charged supercell scaling factor was not considered [$\alpha = 1$ for Eq. (S8) in the [supplementary material](#)]. The supercell volume was $\tilde{V} = 1100.75 \text{ \AA}^3$. Carrier effective mass values used were $m_e^* = 0.2m_e$ for electrons^{39,51} and $m_h^* = 0.8m_e$ for holes,⁵² where m_e is the electron rest mass. For the density of states carrier masses used in emission coefficient calculations, $m_c^* = 0.2m_e$ for the conduction band^{52,53} and $m_v^* = 1.5m_e$ for the valence band.^{54,55} The value of the static dielectric constant used was 11.71, computed from perturbation theory (IBRION = 8) calculations of a GaN primitive unit cell in VASP. These calculations were set up with `vise`⁴⁴ and used the Perdew–Burke–Ernzerhof (PBE) exchange–correlation functional⁸³ along with a $12 \times 12 \times 6$ Γ -centered k -point mesh. Configurational degeneracies were determined based on symmetry by visualizing the structures and comparing nearest neighbor distances. Spin degeneracies were gathered using `doped`⁴⁵ and halved. Capture pathway degeneracies were approximated with Eq. (S6) in the [supplementary material](#) and verified manually.

A sharp drop-off to essentially zero within a single temperature increment occurs for many of the capture coefficients below 60 K, likely a computational artifact since experimental evaluations of low-temperature capture coefficients yield comparable values to those closer to room temperature.^{28,56} Carrier capture coefficient plots are, therefore, presented starting at 60 K. Temperature was considered in two primary methods: filling higher vibrational levels and the Sommerfeld parameter. Increased temperature leading to nonzero occupation ($w_m \neq 0$) of a given vibrational level χ_{im} leads to additional nonzero terms in the summation across all levels in the calculation of the capture coefficient via Eq. (S3) in the [supplementary material](#), thereby increasing the coefficient value.^{5,57} At finite temperatures, transfer can occur at any vibrational level but becomes notably more efficient near ΔE_b .⁴⁹ At low and intermediate temperatures, phonon-assisted tunneling may occur with a lower effective capture barrier, $\Delta E_b'$. These values were determined by fitting the high-temperature unscaled capture coefficient values to Eq. (S9) in the [supplementary material](#).¹³ This exponential form is related to the relationship for PL intensity as well as temperature quenching of luminescence bands via the Mott–Seitz mechanism.^{49,56} Inclusion of a temperature-independent contribution (tunneling) yields lower activation energy required to describe the higher temperature regime associated with MPE.

The Sommerfeld parameter, given by Eq. (S7) in the [supplementary material](#), accounts for Coulombic interaction between the charge carrier and a charged defect.^{50,58} The temperature dependence of the Sommerfeld parameter is dictated by whether the interaction is attractive or repulsive. Attractive interactions have a multiplicative effect but decrease as a function of temperature. Repulsive interactions decrease the capture coefficient value with a lessening effect at increasing temperatures. An example of this for carrier interactions with a negatively charged C_N defect is presented in Fig. 1.

Additional physical materials phenomena should be incorporated to accurately account for temperature effects. As temperature increases, the electronic bandgap of the bulk material is expected

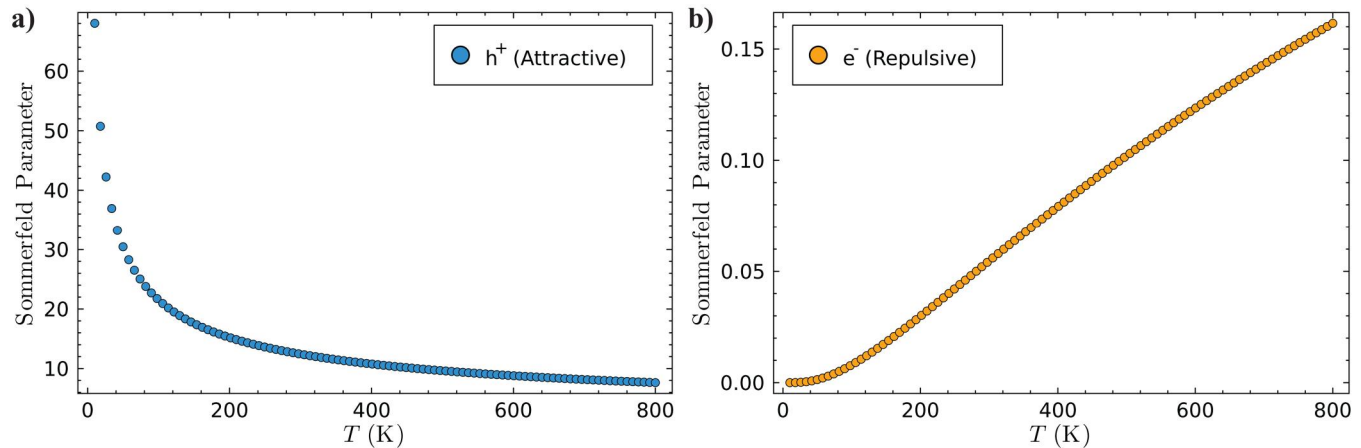


FIG. 1. Example Sommerfeld parameter for singly charged (a) attractive interactions and (b) repulsive interactions. In this case, the defect considered is negatively charged. The interaction types would change for the same charge carriers with a positively charged defect, with a corresponding shift in the value based on the effective carrier mass. For defects with greater charge, the trends remain, but the magnitude of the parameter increases for attractive interactions and decreases for repulsive interactions.

to decrease due to thermal expansion of the lattice as well as electron-phonon interactions.^{59–61} The VBM and CBM are both expected to decrease, with the CBM decreasing at a higher rate. Since the CTL values are determined relative to the VBM, ΔE values used in Eq. (S3) of the [supplementary material](#) will shift accordingly. Small shifts in these CTL values can correspond to significant changes in the capture coefficient. Full calculations of the changes in the bandgap with temperature were not considered here, but the coefficient sensitivity to ΔE is included to demonstrate the effect. Calculations of the expected decrease in bandgap due to the thermal expansion only are included in [Fig. 2](#). Single-point energy calculations were performed with lattice parameters adjusted by the linear coefficients of thermal expansion⁶² for 300, 600, and 900 K. A linear fit was performed, yielding

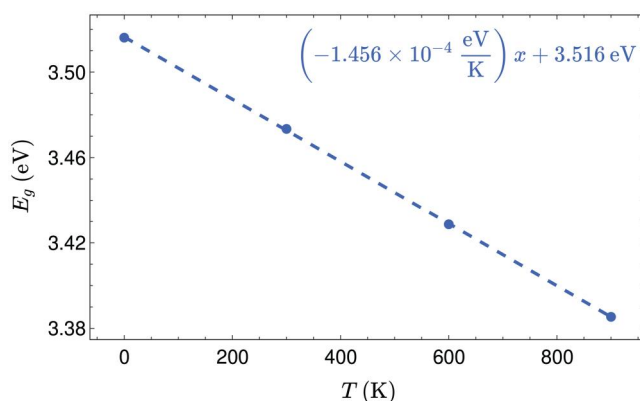


FIG. 2. Decrease in GaN bandgap energy due to lattice thermal expansion with the corresponding linear relationship.

a bandgap temperature coefficient $dE_g/dT = -1.456 \times 10^{-4} \text{ eV/K}$. This accounts for roughly a quarter of the overall shift expected, $dE_g/dT = -6.0 \times 10^{-4} \text{ eV/K}$.³⁹

A high-throughput methodology was developed and included in the `radDefects`⁶³ code. This methodology builds upon `CarrierCapture.jl`⁴⁷ and `nonrad`⁵⁰ by providing robust functions to more efficiently generate VASP input files and perform calculations. Potential energy curve data and calculation parameters, such as the electron-phonon coupling matrix elements and CTL values, are gathered to be read into templated Jupyter notebooks provided for the carrier capture analysis. Similar to some previous work,^{64,65} here, the higher charge state is always considered to be the initial state for consistency in the calculation procedure. This was confirmed to be an equivalent representation as to if the configuration coordinate diagram is defined in reverse. ΔE always defines the distance between the equilibrium energies for the hole capture process, and the electron-phonon coupling matrix elements between the defect state and valence (conduction) band edges are still used for the hole (electron) capture calculation.

III. RESULTS AND DISCUSSION

A. Formation energies and charge transition levels

Quantification of defect properties, specifically formation energies and charge transition levels, grant information about the impact of these defects on the electronic behavior. During the annealing process, charge transfer occurs, enabling the defects to relax into more energetically favorable configurations through localized charge on the defect. *Ab initio* calculations of eight unique defects in GaN were performed. Six native defects were considered: both vacancies (V_{Ga} and V_{N}), interstitials (Ga_i and N_i), and antisites (Ga_{N} and N_{Ga}). Two potential sites were considered

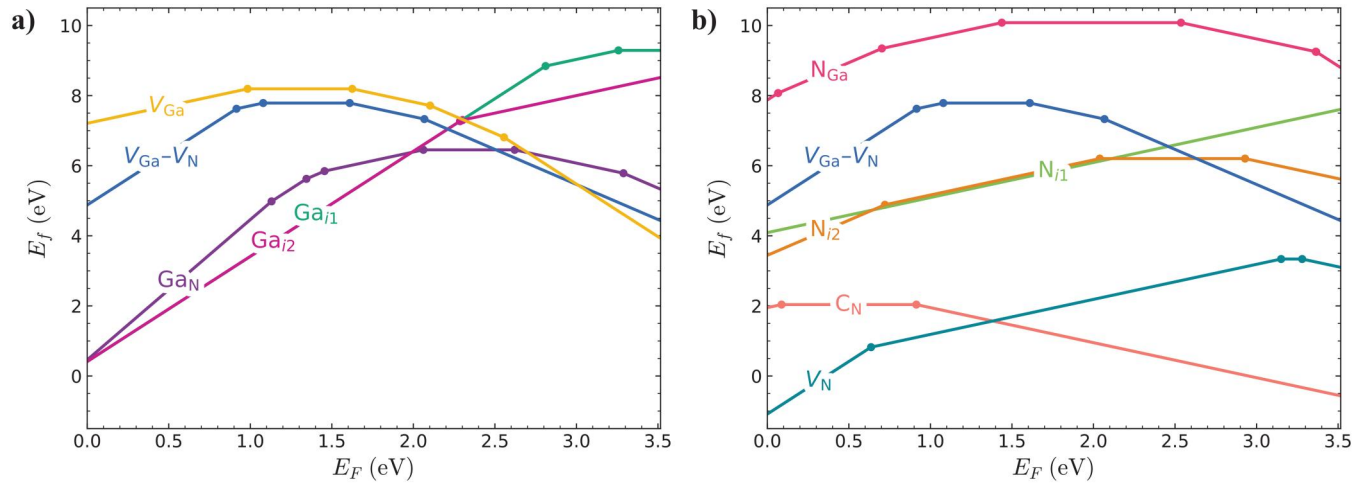


FIG. 3. Formation energies for GaN defects in a Ga-rich environment as a function of Fermi level. Defects include native defects (Ga/N vacancies, interstitials, and anti-sites), C substitutional at a N site, and a divacancy (Ga and N vacancy along the axial direction). (a) Ga-related defects. (b) N-related defects.

for each type of interstitial. In addition to these, two other defects were modeled. Due to its prevalence from radiation damage, a Ga-N divacancy ($V_{\text{Ga}}-V_{\text{N}}$) was also considered.⁶⁶ A common deep acceptor impurity in GaN, a C substitutional atom on a N site (C_{N}), was included in these calculations because of its previous usage as a benchmark for nonradiative carrier capture studies.^{13,67}

Results from Fig. 3 compare agreeably with previous results.^{68–70} A detailed comparison of charge transition levels is presented in Table I. In most cases, where the same transition levels were identified, the values match within a couple tenths of an eV. Several transitions exhibit negative U behavior, typically indicating large configurational changes across charge states.⁴⁸ For these, crossover values for the intermediate charges are still reported where available.

Previous work by Edwards *et al.* considered the nitrogen interstitial in zincblende GaN and identified a split interstitial configuration along $\langle 100 \rangle$ as the most stable for nearly all charge states, shifting to $\langle 110 \rangle$ for the -1 charge state.⁷² Gao *et al.* identified both $\langle 01\bar{1} \rangle$ and $\langle 100 \rangle$ configurations for the split N interstitial in wurtzite GaN.⁷¹ Here, the N interstitial relaxed into two split interstitial configurations. N_{i1} partially occupies the octahedral site along $\langle 01\bar{1} \rangle$, remaining in the $+1$ charge state for all Fermi levels. Neutral and positive charge states for N_{i2} were found to be approximately along the $\langle 212 \rangle$ direction, with one N roughly in-plane with the neighboring Ga atoms. This direction is aligned with the long diagonal of the hexagonal structure, in line with a N 6.89 Å away across two open interstitials. The second N is 1.34 Å away in the neutral state. For increasingly positive charge states, the first N shifts further into the open interstitial area and pushes the neighboring Ga atoms outward. The separation between N atoms also decreases. The N-N distance is 0.05 Å longer than in Edwards *et al.*⁷² for the neutral state and 0.05 Å shorter for $+3$.

In the -1 state, N_{i2} reconfigures along the $\langle \bar{1}22 \rangle$ direction, with a distance of 1.47 Å between the N atoms (0.07 Å shorter than in Edwards *et al.*). This direction is toward a second nearest

neighbor Ga across the short diagonal of the open interstitial. It appears similar to the neutral configuration identified by von Bardeleben *et al.*⁷³ as responsible for their EPR signal corresponding to a radiation-induced paramagnetic defect, which they validated using HSE calculations. Von Bardeleben *et al.* consider n-type GaN with a Fermi level 35 meV below the CBM that shifts to 1.0 eV below the CBM for proton fluences above $1 \times 10^{-16} \text{ cm}^{-2}$. They found the $(0/-1)$ and $(+1/0)$ transitions of split N_i to occur 1.01 and 1.45 eV below the CBM, respectively. The present $(0/-1)$ transition value is closer to the CBM, just 0.59 eV below it, but the $(+1/0)$ value of 1.48 eV below is quite similar. These values still support the conclusion by von Bardeleben *et al.*⁷³ that the N split interstitial will be in the neutral state at the pinned Fermi level.

The Ga interstitial favors the octahedral site for all Fermi levels, particularly for increasingly positive charge states, identified

TABLE I. Thermodynamic charge transition levels, $\Delta E(q_1/q_2)$, for native defects in GaN. Values are presented against previous computational works with all values relative to the VBM.

Defect	q_1/q_2	$\Delta E(q_1/q_2)$ (eV)			
		This work	Gao <i>et al.</i> ⁷¹	Lyons and Van de Walle ⁶⁹	Diallo and Demchenko ⁶⁸
V_{Ga}	$(+1/0)$	0.98	0.68	0.97	0.94
	$(0/-1)$	1.63	1.55	1.68	1.73
	$(-1/-2)$	2.10	1.98	2.33	1.87
	$(-2/-3)$	2.56	2.42	2.80	2.34
V_{N}	$(+3/+1)^a$	0.64	0.90	0.47	0.54
	$(+3/+2)$	0.66	... ^c	... ^c	0.61

TABLE I. (Continued.)

Defect	q_1/q_2	$\Delta E (q_1/q_2)$ (eV)			
		This work	Gao <i>et al.</i> ⁷¹	Lyons and Van de Walle ⁶⁹	Diallo and Demchenko ⁶⁸
V_{Ga}^- V_{N}	(+2/+1)	0.61	... ^c	... ^c	0.47
	(+1/0)	3.15	3.10	3.27	3.49 ^b
	(0/-1)	3.28	... ^c	3.55 ^b	... ^c
	(+3/+1) ^a	0.92	N/A ^d	N/A ^d	0.74
	(+3/+2)	0.94			0.81
	(+2/+1)	0.89			0.68
	(+1/0)	1.08			0.98
	(0/-1)	1.61			1.48
	(-1/-2)	2.07			1.95
	(+3/+1) ^a	2.81	2.04	N/A ^d	N/A ^d
Ga_{il}	(+3/+2)	3.26	... ^c		
	(+2/+1)	2.36	... ^c		
	(+1/0)	3.26	3.31		
	(+3/+2)	2.29	2.33	2.09	2.33
Ga_{i2}	(+2/+1)	2.30	2.58	2.57	2.64
	(+1/0)	... ^c	... ^c	... ^c	3.49 ^b
	+1	... ^c	N/A ^d	N/A ^d	N/A ^d
N_{i1}	(+3/+2)	... ^c	0.75	0.02	0.22
	(+2/+1)	0.72	1.48	0.54	0.51
	(+1/0)	2.04	2.06	2.34	2.45
	(0/-1)	2.93	2.65	2.52	3.02
N_{i2}	(+3/+2)	... ^c	0.75	0.02	0.22
	(+2/+1)	0.72	1.48	0.54	0.51
	(+1/0)	2.04	2.06	2.34	2.45
	(0/-1)	2.93	2.65	2.52	3.02
	(+4/+3)	1.13	... ^c	1.20	0.52
	(+3/+2)	1.34	1.58	1.39	1.24
Ga_{N}	(+2/+1)	1.46	1.85	1.46	1.50
	(+1/0)	2.06	2.20	2.32	2.42
	(0/-1)	2.62	2.75	2.76	2.99
	(-1/-2)	3.29	3.10	... ^c	3.44
	(+3/+2)	0.07	0.69	0.41	... ^c
	(+2/0) ^a	... ^c	... ^c	... ^c	2.40
N_{Ga}	(+2/+1)	0.70	1.31	1.47	2.76
	(+1/0)	1.44	1.50	1.70	2.03
	(0/-1)	2.54	2.96	2.44	... ^c
	(-1/-2)	3.36	3.37	... ^c	... ^c
	(-2/-3)	3.37	... ^c	... ^c	... ^c

^aCTL exhibits $-U$ behavior.^bShallow CTL resonant with the CBM.^cCTL not identified.^dDefect not considered.

as Ga_{i2} here. $\text{Ga}_{\text{i}}^{+3}$ is predominantly in the center of the octahedral interstitial, matching the structures shown in previous work^{68,71} well. Ga_{i} is more likely to relax into the tetrahedral site at lower charges, becoming competitive (<1 eV higher) near the CBM. Because of this, Ga_{i1} occupies the octahedral site in the +3 state and the tetrahedral site in the +1 and neutral states. The

tetrahedral Ga is in-plane with the N atoms, pushing the closest neighboring N atom to be approximately in line with the neighboring lattice Ga along the a -axis. The $-U$ behavior for the (+3/+1) transition arises when moving from the octahedral site to the tetrahedral site. This and the (+1/0) near-CBM transition match qualitatively with previous results for the tetrahedral configuration.⁷¹ The 0.77 eV deviation from their reported (+3/+1) CTL value is most likely due to structural differences. Gao *et al.*⁷¹ detailed an increased distance between Ga_{i} and the neighboring N atoms compared to the length of Ga-N bonds in perfect GaN. Dissimilarly, the present Ga_{i1} pushes one of the neighboring N atoms off of its lattice site, forming a bond with reduced length (1.90 Å).

Results presented for N_{Ga} are qualitatively similar to both Gao *et al.* as well as Lyons and Van de Walle.^{69,71} The (-1/-2) and (-2/-3) transitions are within close proximity to both each other (~0.01 eV) and to the CBM (<0.2 eV). Both of these energy differences are within the expected error range for CTL values, meaning it is reasonable for either one or both of these transitions to not occur in these previous studies. Some of the values deviate by up to 0.77 eV, namely, the (+2/+1) transition, likely the result of structural variance. Gao *et al.*⁷¹ discusses the complexity of the structural distortion for N_{Ga} , stating that it tightly bonds with a neighboring N atom and shifts off-site. N_{Ga}^0 specifically moves upward along the c -axis to form a 1.22 Å bond with the axial N. Present results show N_{Ga}^0 forming a 1.17 Å bond with a neighboring basal plane N. Much more significant differences were found by Diallo and Demchenko, identifying $-U$ behavior in the sole transition, (+2/0).⁶⁸ They report that $\text{N}_{\text{Ga}}^{+2}$ remains at the lattice site with a major reduction in neighboring N bond length. Here, the $\text{N}_{\text{Ga}}^{+2}$ configuration is very similar to that of the neutral, with a further reduction in bond length to 1.15 Å.

B. Benchmark: C_{N}

Several previous works on carrier capture in GaN focus on C_{N} because they are commonplace and their connection to yellow and blue luminescence.^{13,19–22,29–31,74} A summary of the previous DFT calculations for the C_{N} defect can be found in several review and seminal papers.^{13,69,75–77} C_{N} was shown to be a deep acceptor with a (-1/0) transition near 1 eV above the VBM, related to yellow luminescence. Additionally, a (0/+1) transition exists closer to the VBM, which was linked to blue luminescence.²⁸ CTL values for these, determined computationally, are reported in Table II. The ionization energy for the (-1/0) transition was determined experimentally by PL to be 0.916 ± 0.003 eV,³¹ close to the deep-level transient spectroscopy (DLTS) value of 0.88 eV.⁷⁸ For the (0/+1) transition, the PL value of 0.33 ± 0.01 eV was found,³⁰ also close to the 0.29 eV value found from DLTS.⁷⁸ The experimental (-1/0) transition is very close to the value calculated presently, whereas the (0/+1) transition is calculated to be ~0.25 eV lower. Parameters for the hole capture calculations are shown in Table III: the charge transition level (ΔE), the coordinate difference between equilibrium configurations (ΔQ), classical capture barriers (ΔE_{b}), effective capture barriers ($\Delta E_{\text{b}}^{\text{eff}}$), capture pathway degeneracy (g), and electron-phonon coupling matrix elements ($\dot{W}_{ij}$).

TABLE II. Thermodynamic charge transition levels, ΔE (q_1/q_2), for C_N in GaN. Values are presented against previous computational works with all values relative to the VBM.

q_1/q_2	ΔE (q_1/q_2) (eV)				
	This work	Zhao <i>et al.</i> ²²	Alkauskas <i>et al.</i> ¹³	Lyons <i>et al.</i> ⁶⁷	Demchenko <i>et al.</i> ²¹
(+1/0)	0.09	0.21	... ^a	0.35	0.43
(0/−1)	0.91	1.06	1.02	0.90	1.09

^aCTL not considered.

1. C_N (−1/0)

Figure 4 shows the nonradiative hole capture process associated with the C_N^{-1} to C_N^0 transition. The −1 state configuration corresponds to C occupying the N site with equivalent Ga–C bond distances within 0.01 Å. The bonds lengthen unequally during hole capture into the neutral configuration, where three of the bonds increase by approximately 2% and the fourth by roughly 5%. The CCD in Fig. 4(a) shows the potential energy curve for each charge state as relatively harmonic. The classical transition barrier was found from the crossing point in the two curves relative to the minimum energy of the initial state (C_N^{-1}), yielding $\Delta E_b = 0.66$ eV. Equation S9 in the [supplementary material](#) was used to extract an effective hole capture barrier. The unscaled capture coefficient curve and the fitted curve are shown in Fig. 5. The value of the effective barrier is $\Delta E'_b = 0.25$ eV, similar to the 0.23 eV found by Alkauskas *et al.*¹³ The electron–phonon coupling matrix element, 9.1×10^{-2} eV/amu^{1/2} Å, was determined using the neutral equilibrium configuration as the reference state.

The scaled hole capture coefficient C_p is presented as the solid blue line in Fig. 4(b) and is reasonably high in value. The hole capture coefficient increases for most of the considered temperature range, both due to the filling of higher vibrational levels and the attractive Sommerfeld parameter [as in Fig. 1(a)]. The value at 300 K is 3.2×10^{-9} cm³/s, which compares closely with the value from Alkauskas *et al.*,¹³ 3.1×10^{-9} cm³/s, shown by the red diamond at 300 K. At 390 K, the coefficient is 7.0×10^{-9} cm³/s, which also compares well with the more recent results from Zhao *et al.*,²² 1×10^{-9} cm³/s.

Dashed blue lines in Fig. 4(b) bound the hole capture curve to show the coefficient's sensitivity to the CTL value by recalculating the capture coefficient with ΔE adjusted by ± 0.05 eV. An increase by 0.05 eV leads to a coefficient of 4.9×10^{-10} cm³/s at 300 K, while a decrease of 0.05 eV yields 2.0×10^{-8} cm³/s. This highlights that a small change in the CTL may result in orders of magnitude difference in the capture coefficient, depending on the CCD. At 600 K, Alkauskas *et al.* applied a simple finite

temperature approximation, decreasing the bandgap by 0.26 eV and attributing half of this to each of the VBM and the CBM. This effectively shifted the CTL by 0.13 eV, which led to their value of 3.1×10^{-7} cm³/s (red diamond at 600 K) as well as the presently approximated value of 9.0×10^{-7} cm³/s (turquoise square). Photoluminescence measurements from Reshchikov *et al.*³¹ corresponding to the yellow luminescence band YL1 were determined to be $(3.7 \pm 1.6) \times 10^{-7}$ at 100 K and 3×10^{-7} cm³/s from 450 to 700 K, shown as the dashed yellow lines with “X” markers.

Electron capture coefficients were calculated but are only shown in Fig. S15 of the [supplementary material](#) due to the low value at all temperatures. For example, at 390 K, the value was 2.4×10^{-24} cm³/s, about three orders of magnitude below that from Zhao *et al.*,²² 3×10^{-21} cm³/s. Several methodological differences exist that may lead to this discrepancy. Notably, norm-conserving pseudopotentials⁷⁹ were used with a cutoff energy of 120 Ry by Zhao *et al.* as opposed to the PAW potentials with a 400 eV cutoff here. The FNV^{80,81} finite-size correction scheme was also used rather than the eFNV scheme. These differences lead to the variation of CTL values in Table II. Zhao *et al.* constructed CCDs and computed coefficients with nonrad assuming harmonic potential energy curves, in contrast with spline fits from CarrierCapture.jl used here.

2. C_N (0/+1)

Results for the (0/+1) transition of C_N are presented in Fig. 6. The Ga–C bonds from the neutral state all lengthen further, with the +1 configuration having two equivalent distances and two extended, inequivalent bonds. The hole capture classical barrier was observed in Fig. 6(a) to be 0.06 eV. The capture coefficient exhibits mild temperature dependence as expected because of the small classical barrier. The previous coefficient value from Zhao *et al.*²² at 390 K was 4×10^{-6} cm³/s, about three orders of magnitude greater than the value calculated here, 3.8×10^{-9} cm³/s. This discrepancy is indicative of the sensitivity of the capture coefficient to small differences in the energy landscape due to methodology differences.

Experimental work by Reshchikov^{27,31} connects the blue luminescence band, BL_C, with a hole capture coefficient of approximately 1×10^{-10} cm³/s to this transition. As mentioned above, the (0/+1) transition has been experimentally established to occur at 0.3 eV above the VBM.^{27,31,78} The electron capture coefficient (390 K) is 6×10^{-19} cm³/s from Zhao *et al.* and 1.5×10^{-21} cm³/s here, much smaller than the hole capture coefficient in both cases.

C. Carrier capture coefficients

Analysis of individual defects and associated carrier capture calculations is presented below. A summary of all those computed

TABLE III. Nonradiative carrier capture calculation parameters for C_N in GaN.

q_1/q_2	ΔE (eV)	ΔQ (amu ^{1/2} Å)	Carrier	ΔE_b (eV)	$\Delta E'_b$ (eV)	g	$\tilde{W}_{if}$ (eV/amu ^{1/2} Å)
(−1/0)	0.91	1.55	h^+	0.66	0.25	4	9.1×10^{-2}
(0/+1)	0.09	1.69	h^+	0.06	0.05	1	4.8×10^{-2}

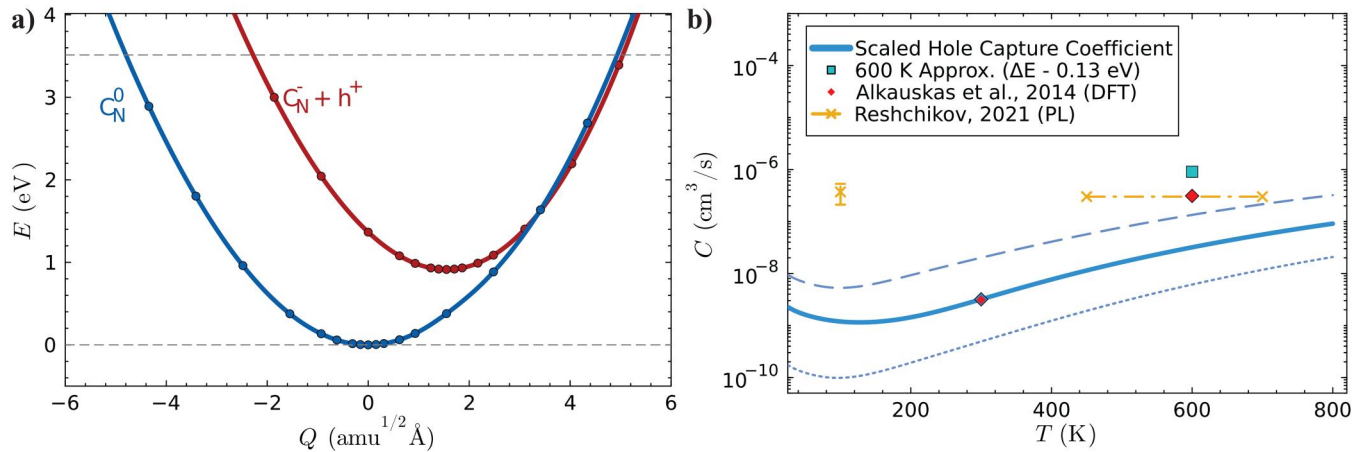


FIG. 4. (a) Configuration coordinate diagram for the C_N ($-1/0$) transition. (b) Scaled hole capture coefficient for C_N ($-1/0$) and sensitivity curves for $\Delta E \pm 0.05$ eV (dotted line is $+0.05$ eV and dashed line is -0.05 eV). A high-temperature approximation with $\Delta E = 0.13$ eV is shown to compare with previous computational and photoluminescence (YL1) results.

is found in Fig. 7 where the CTLs are plotted throughout the bandgap at the Fermi level they appear at for each defect. The color of each CTL indicates its relative carrier capture coefficient value at 300 K compared to all others (the same scale is used between hole and electron capture). The colorscale used corresponds dark green with relatively high carrier capture coefficients and indigo with relatively low carrier capture coefficients. High hole capture values in Fig. 7(a) are primarily concentrated toward the VBM, while the more significant electron capture values in Fig. 7(b) are closer to the CBM. The three highest capture coefficients overall are all associated with electron capture by Ga_i ($+3/+2$) and ($+2/+1$) octahedral transitions, and then the ($+1/0$) tetrahedral transition. These are

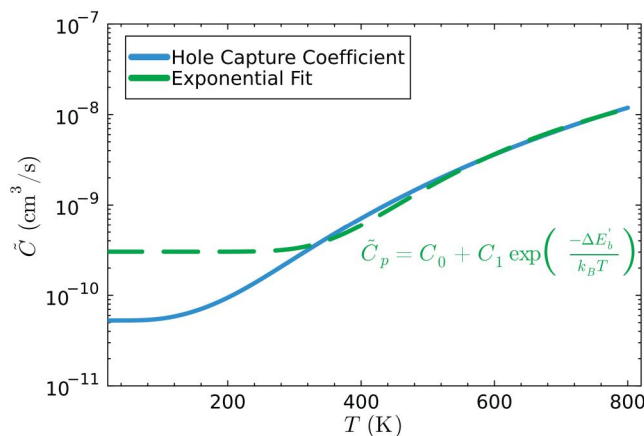


FIG. 5. C_N ($-1/0$) unscaled hole capture coefficient with an exponential fit. The fitting parameters were determined to be $C_0 = 3.04 \times 10^{-10}$ cm³/s, $C_1 = 4.36 \times 10^{-7}$ cm³/s, and $\Delta E_b = 0.25$ eV.

followed by ($+1/0$) electron capture for V_N and ($0/+1$) hole capture for V_{Ga} . A ($+3/+1$) multi-charge transition appears for V_N , $V_{Ga}-V_N$, and Ga_{i1} . Due to the additional complexities of the inclusion of metastable states and interactions among multiple charge carriers during the capture process, capture coefficients were not computed for these three transitions.

Nonradiative capture is considered dominant for transitions with capture coefficients above 10^{-10} cm³/s,¹³ and therefore, these processes are expected to more significantly impact device performance. Seven transitions for the native point defects and the divacancy meet these criteria at 300 K. Almost all of these processes are occurring between neutral and singly charged states. Only one of these transitions, the ($0/+1$) transition for V_{Ga} , has a greater hole capture coefficient at room temperature, identifiable by the charge transition level closer to the VBM. Table IV provides calculation parameters for each of these transitions with the same definitions as Table III. In most cases, an effective capture barrier was able to be determined from an exponential fit of the unscaled capture coefficients. Further details are provided below for the coefficients above 10^{-10} cm³/s, and the effective barrier fits and remaining coefficients are included in Sec. S B of the supplementary material. Table SI in the supplementary material includes values of all the capture coefficients at 60, 300, and 800 K with all parameters given in Table SII of the supplementary material. These values can be considered alongside the relative proportion of defects produced via displacement damage in GaN to understand the overall expected impact.⁶⁶

1. Ga_{i2} ($+3/+2$) and ($+2/+1$)

Octahedral Ga_{i2} has two CTLs, very close in value to each other and just over 1 eV below the CBM. Both transitions have substantial electron capture coefficients at all temperatures considered. Potential energy curves involved in the transition

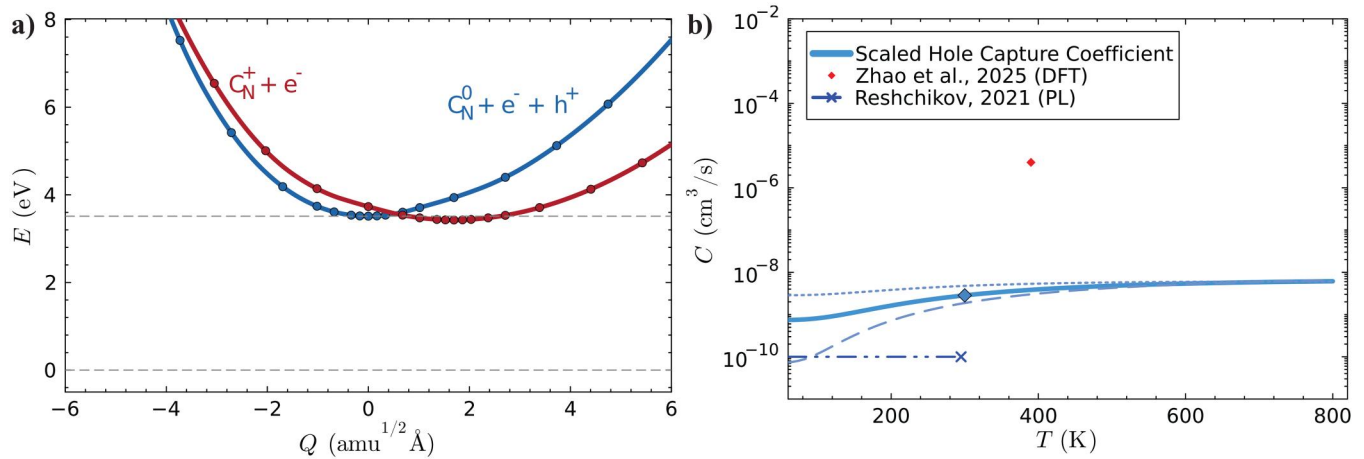


FIG. 6. (a) Configuration coordinate diagram for C_N (0/+1). (b) Scaled hole capture coefficient for C_N (0/+1) and sensitivity curves for $\Delta E \pm 0.05$ eV (dotted line is +0.05 eV and dashed line is -0.05 eV). Low- and room-temperature computational and photoluminescence (BL_C) results from previous work are included.

are nearly harmonic, shown in Fig. 8(a). The electron capture barrier is extremely low: 0.03 eV. The electron capture coefficient in Fig. 8(b) shows that the value always remains above 10^{-5} cm³/s, despite a slight decrease over increasing temperature. This is also true for the CTL sensitivity bounds, which match the same trend as the base coefficient curve.

The (+2/+1) transition undergoes a much larger configurational shift, with $\Delta Q = 6.62$ amu^{1/2}Å compared to 3.09 amu^{1/2}Å for (+3/+2). Potential energy curves for the +2 and +1 states are significantly anharmonic, with large overlap occurring between them in Fig. 9(a). The overlap begins at just 0.14 eV above the equilibrium energy for Ga_{i2}^{+2} . The electron capture coefficient increases

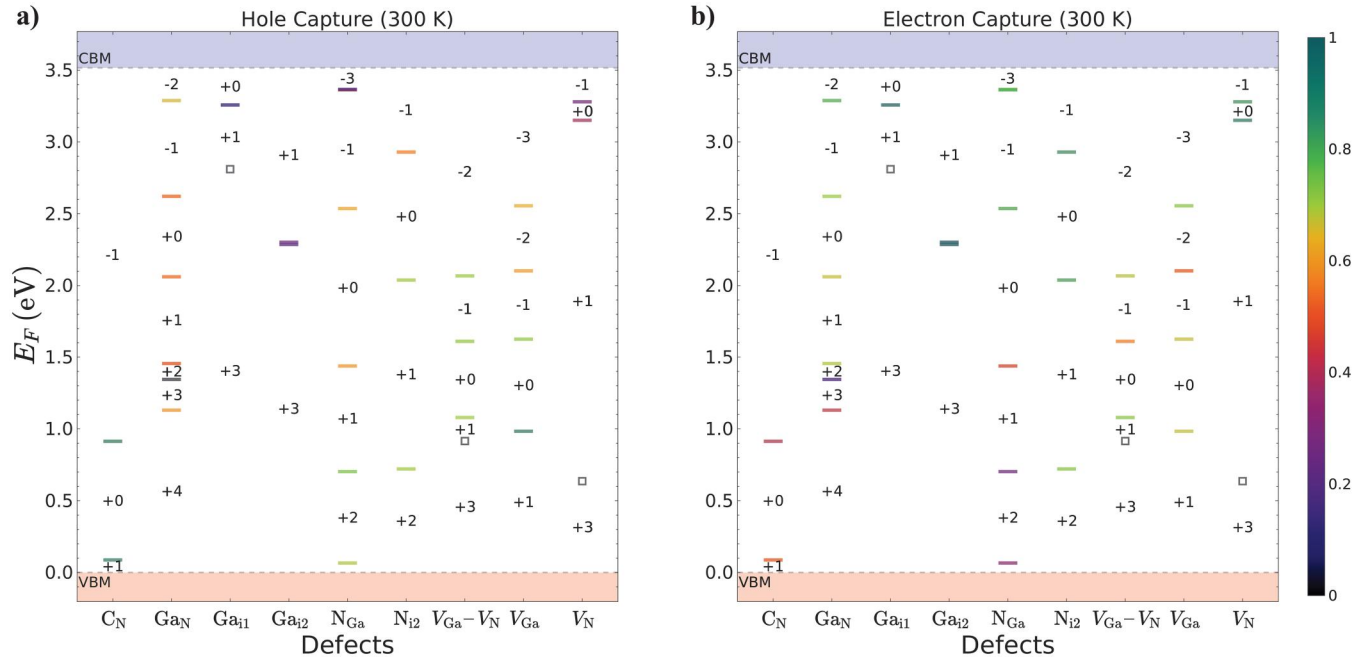


FIG. 7. Charge transition levels for GaN defects in a Ga-rich environment. The colorscale indicates the relative scaled carrier capture coefficient values at 300 K for (a) hole capture and (b) electron capture. Multi-charge transitions, i.e., the (+3/+1) for V_N , $V_{Ga}-V_N$, and Ga_{i1} , were not considered here and are therefore displayed as open markers. The +2 state for Ga_{i2} and the +3 and -2 states for N_{Ga} are not labeled, only existing for <0.08 eV.

TABLE IV. Nonradiative carrier capture calculation parameters for defect transitions with capture coefficients above 10^{-10} cm³/s at 300 K in GaN.

Defect	q_1/q_2	ΔE (eV)	ΔQ (amu ^{1/2} Å)	Carrier	ΔE_b (eV)	$\Delta E'_b$ (eV)	g	$\tilde{W}_{if}$ (eV/amu ^{1/2} Å)
Ga _{i2}	(+3/+2)	2.29	3.09	e ⁻	0.03	0.02	1	1.9×10^{-1}
Ga _{i2}	(+2/+1)	2.30	6.62	e ⁻	0.14	0.09	1.5	1.0×10^{-1}
Ga _{i1}	(+1/0)	3.26	3.29	e ⁻	0.07	0.05	1	1.2×10^{-1}
V _N	(+1/0)	3.15	2.88	e ⁻	0.03	0.02	1	4.7×10^{-3}
V _{Ga}	(0/+1)	0.98	1.97	h ⁺	0.09	0.07	0.5	1.0×10^{-1}
N _{i2}	(0/-1)	2.93	3.51	e ⁻	0.01	... ^a	0.5	1.3×10^{-1}
V _N	(0/-1)	3.28	3.03	e ⁻	0.07	0.06	1	8.7×10^{-3}

^aCannot obtain good exponential fit.

less than an order of magnitude across the temperature range considered, seen in Fig. 9(b). At 60 K, the electron capture coefficient is 5.8×10^{-6} cm³/s, and at 800 K, the value is 4.9×10^{-5} cm³/s. CTL sensitivity bounds follow a similar trend but encompass a wide range of coefficient values.

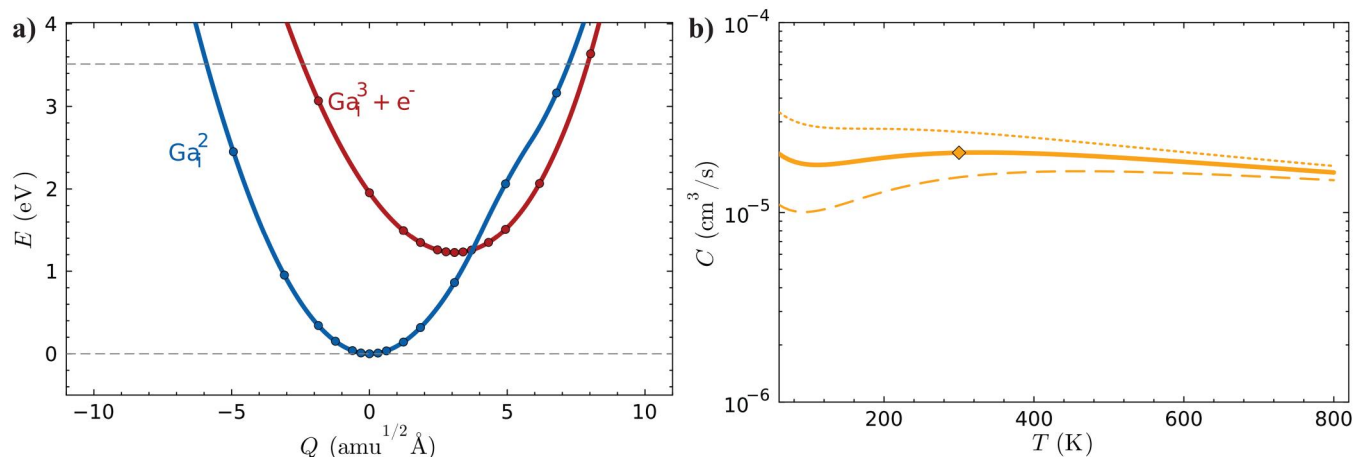
2. Ga_{i1} (+1/0)

Ga_{i1} has two CTLs, both within 1 eV of the CBM. The lower of the two is the multi-charge (+3/+1) transition at 2.81 eV, indicating the defect exists in the +3 charge state for intrinsic and p-type GaN. The (+1/0) transition occurs at 3.26 eV, and the electron capture coefficient for which is exceptionally large at 300 K, as seen in Fig. 10. In Fig. 10(a), the potential energy curve for the +1 state is highly parabolic. The neutral state curve becomes non-parabolic after extrapolating past the equilibrium configuration of the positive state. An electron capture barrier of $\Delta E_b = 0.07$ eV is aligned with how shallow the CTL is relative to the CBM. The electron capture coefficient, displayed in Fig. 10(b), increases from just above 10^{-8} cm³/s to a near-saturation point around 10^{-6} cm³/s

by 300 K. The CTL sensitivity bounds mirror the base coefficient line shape and follow the same trends as C_N (-1/0) hole capture in Fig. 4(b). An increased ΔE raises the classical capture barrier, which results in a lower capture coefficient, and the opposite occurs for a decrease in ΔE .

3. V_N (+1/0) and (0/-1)

Three CTLs were found for V_N, one of which being a multi-charge transition. This (+3/+1) transition is near the VBM (0.64 eV), while the other two [(+1/0) at 3.15 eV and (0/-1) at 3.28 eV] are close to the CBM. V_N is, therefore, expected to be in the +1 charge state for most Fermi energies, and electron capture into the neutral state has a large capture coefficient value at 300 K. In addition, V_N are expected to be abundant in GaN under displacement damage.⁶⁶ The CCD in Fig. 11(a) displays a high-level of harmonicity, with a low classical capture barrier. An electron capture barrier of 0.03 eV was identified. The electron capture coefficient in Fig. 11(b) generally decreases across the range of temperatures. The value at 60 K is 7.5×10^{-9} cm³/s and decreases

**FIG. 8.** (a) Configuration coordinate diagram for Ga_{i2} (+3/+2). (b) Scaled electron capture coefficient for Ga_{i2} (+3/+2) and sensitivity curves for $\Delta E \pm 0.05$ eV (dotted line is +0.05 eV and dashed line is -0.05 eV).

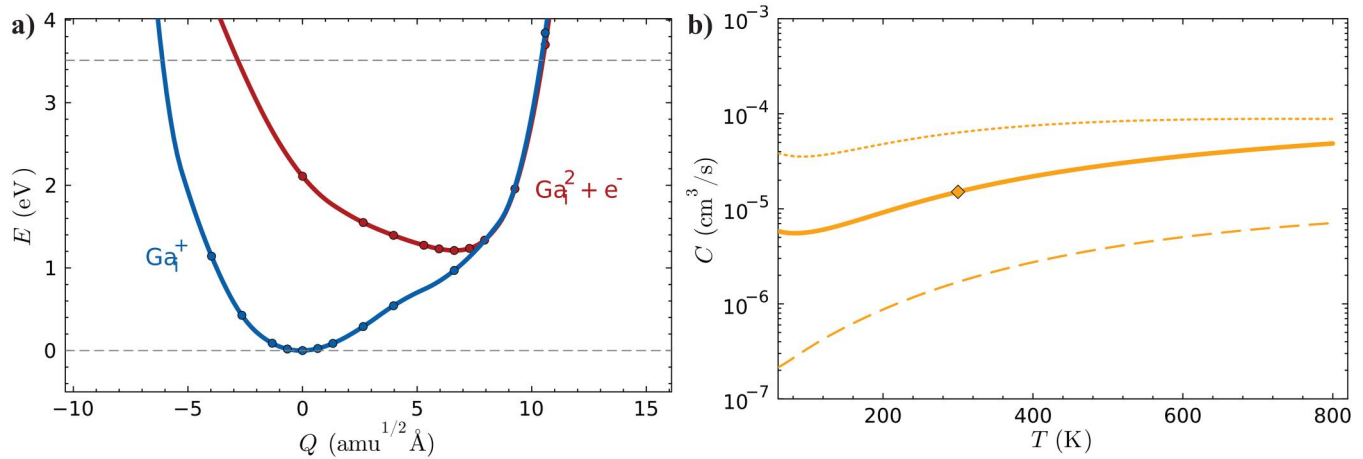


FIG. 9. (a) Configuration coordinate diagram for $\text{Ga}_{22} (+2/+1)$. (b) Scaled electron capture coefficient for $\text{Ga}_{22} (+2/+1)$ and sensitivity curves for $\Delta E \pm 0.05$ eV (dotted line is $+0.05$ eV and dashed line is -0.05 eV).

to $5.1 \times 10^{-9} \text{ cm}^3/\text{s}$ by 800 K. The decrease is mainly due to the increase in free carrier energy making them harder to capture as described by the Sommerfeld parameter (see Fig. 1). Initially, there is a large difference between each of the sensitivity bounds and the base electron capture coefficient, but this difference decreases toward higher temperatures as the bounds converge.

The $V_N (0/-1)$ transition is closer to the CBM than the $(+1/0)$ transition. The present $V_N (0/-1)$ results are shown in Fig. 12, and the electron capture barrier was found via Fig. 12(a) to be 0.07 eV. The coefficient increases two orders of magnitude from 3.3×10^{-12} to $3.9 \times 10^{-10} \text{ cm}^3/\text{s}$. The sensitivity bounds again follow the same trend and converge at high temperatures.

4. $V_{Ga} (0/+1)$

V_{Ga} has four CTLs throughout the bandgap of GaN with the $(0/+1)$ hole capture coefficient being the most important for non-radiative capture. The CTL is in the bottom third of the bandgap at 0.98 eV. Figure 13(a) shows the relatively parabolic potential energy curves along the configuration coordinate between the neutral and $+1$ states. Both states deviate from this when extrapolating past the other state's equilibrium configuration. The hole capture barrier was found to be $\Delta E_b = 0.09$ eV. From 60 to 800 K, the coefficient increases one order of magnitude, seen in Fig. 13(b). The value starts at $1.6 \times 10^{-9} \text{ cm}^3/\text{s}$, reaches $5.2 \times 10^{-9} \text{ cm}^3/\text{s}$ at 300 K, and ends at $1.5 \times 10^{-8} \text{ cm}^3/\text{s}$. As typical,

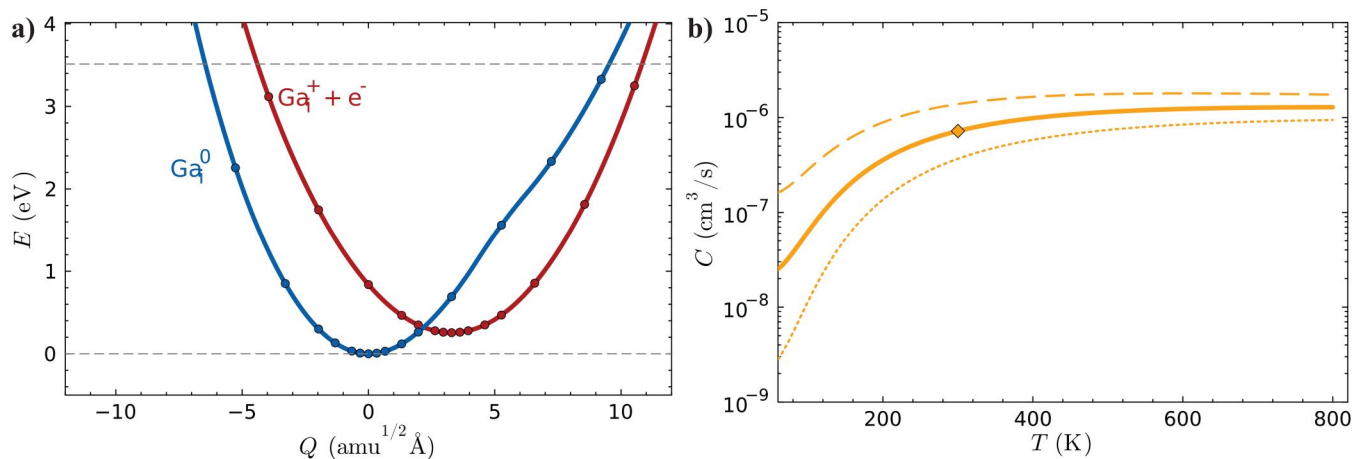


FIG. 10. (a) Configuration coordinate diagram for $\text{Ga}_{11} (+1/0)$. (b) Scaled electron capture coefficient for $\text{Ga}_{11} (+1/0)$ and sensitivity curves for $\Delta E \pm 0.05$ eV (dotted line is $+0.05$ eV and dashed line is -0.05 eV).

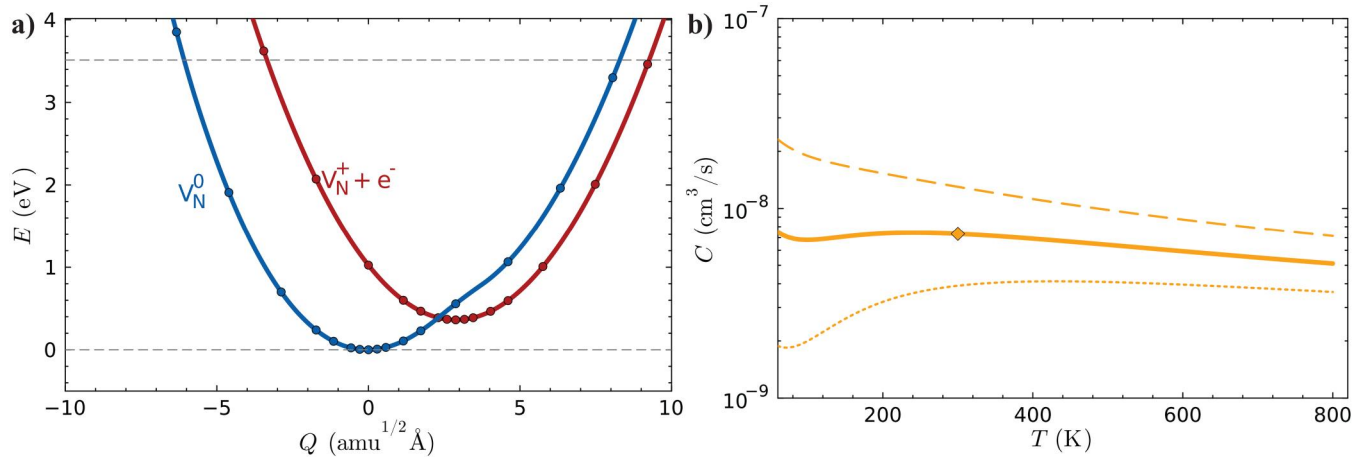


FIG. 11. (a) Configuration coordinate diagram for $V_N (+1/0)$. (b) Scaled electron capture coefficient for $V_N (+1/0)$ and sensitivity curves for $\Delta E \pm 0.05$ eV (dotted line is $+0.05$ eV and dashed line is -0.05 eV).

the sensitivity bounds follow similar increases in coefficient and converge toward the same value as the base case at high temperatures.

5. N_{i2} (0/-1)

Based on simulations and irradiation experiments, N_i defects often have the highest post-irradiation concentration in GaN.^{66,73} Two inequivalent split configurations were identified for the N_i , as detailed in Sec. III A. Only one of these configurations, N_{i2} , has charge transitions within the bandgap. N_{i2} 's (0/-1) transition in Fig. 14(a) yields a small classical barrier for electron capture: 0.01 eV. Due to the small classical barrier, the electron capture

coefficient in Fig. 14(b) is observed to be temperature independent, approximately 4×10^{-10} cm³/s. Because of the temperature independence, an effective barrier could not be determined as a good exponential fit was not found. The coefficient is most sensitive to the overlap of the vibrational wavefunctions between the two quantum oscillators. Small shifts in the CTL can, therefore, lead to exceptionally large variations in the capture coefficient due to improved overlap, as seen by the sensitivity curves in Fig. 14(b). Both of the CTL sensitivity curves are greater than 10^{-9} cm³/s, with a crossing point between them around 120 K. The $\Delta E + 0.05$ eV curve initially starts lower and increases to about 4×10^{-9} cm³/s, while $\Delta E - 0.05$ eV decreases toward 1×10^{-9} cm³/s.

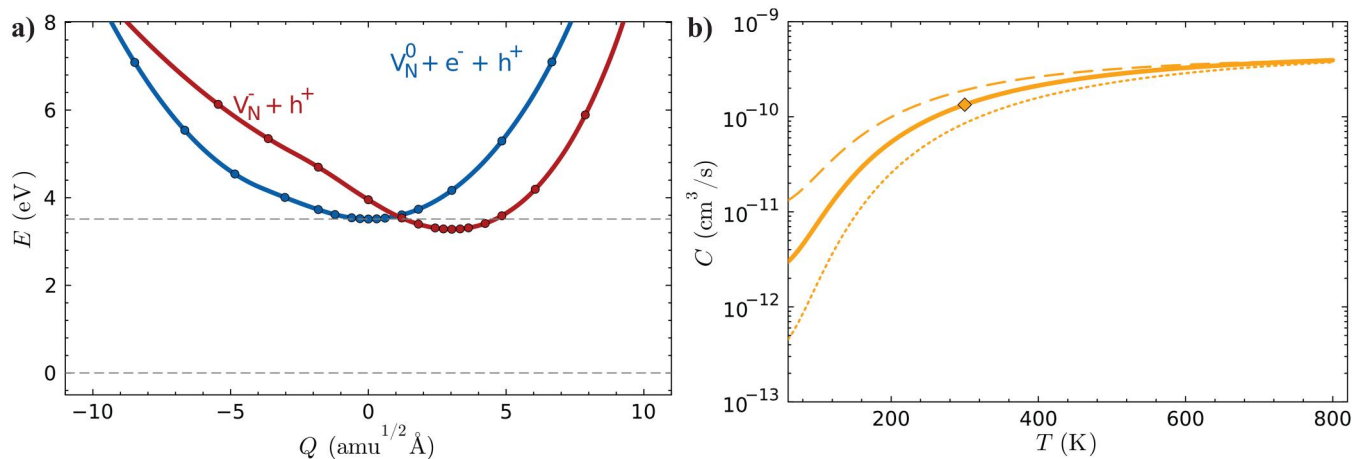


FIG. 12. (a) Configuration coordinate diagram for $V_N (0/-1)$. (b) Scaled electron capture coefficient for $V_N (0/-1)$ and sensitivity curves for $\Delta E \pm 0.05$ eV (dotted line is $+0.05$ eV and dashed line is -0.05 eV).

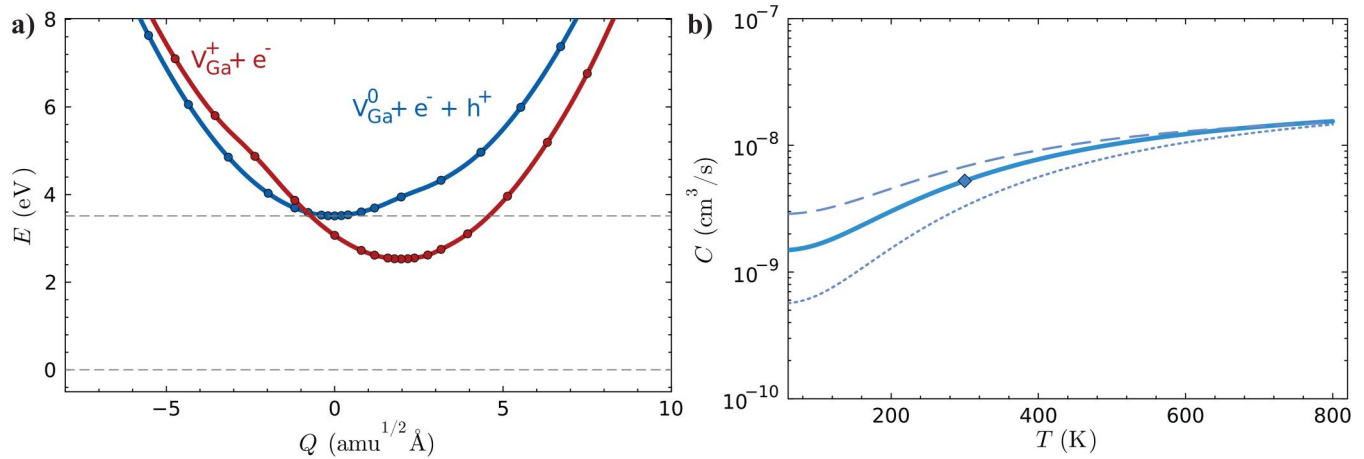


FIG. 13. (a) Configuration coordinate diagram for $V_{\text{Ga}} (0/+1)$. (b) Scaled hole capture coefficient for $V_{\text{Ga}} (0/+1)$ and sensitivity curves for $\Delta E \pm 0.05$ eV (dotted line is $+0.05$ eV and dashed line is -0.05 eV).

D. Device implications

Most coefficients increase a few orders of magnitude at most as a function of temperature, primarily due to the occupation of higher vibrational levels and increased wavefunction overlap. Cases with small barriers often lead to coefficients, which are roughly temperature-independent [e.g., $N_{\text{Iz}} (0/-1)$ electron capture]. When paired with the Sommerfeld parameter, the temperature behavior may be entirely dictated by the interaction type [e.g., $V_{\text{N}} (+1/0)$ electron capture decreasing]. In some instances, much more significant temperature dependence may be observed. This either indicates a highly temperature-activated process [e.g., $N_{\text{Iz}} (+2/+1)$ electron capture increasing over six orders of

magnitude], or it is due to high charge state Sommerfeld parameters that contribute significantly [e.g., $V_{\text{Ga}} (-2/-3)$ electron capture increasing over four orders of magnitude]. Additionally, the expected bandgap decreases because finite temperature electron-phonon interactions shift the CTL value, often increasing the capture coefficient value. While beyond the scope of the present study, a more rigorous approach could be taken to further explore this effect, as done previously by Wickramaratne *et al.*⁵⁹

Through the carrier capture and emission processes, the evolution of defects at different charge states can be solved as a function of time. As done by Wampler *et al.*,²⁹ a system of rate equations can be constructed with various defects to model this

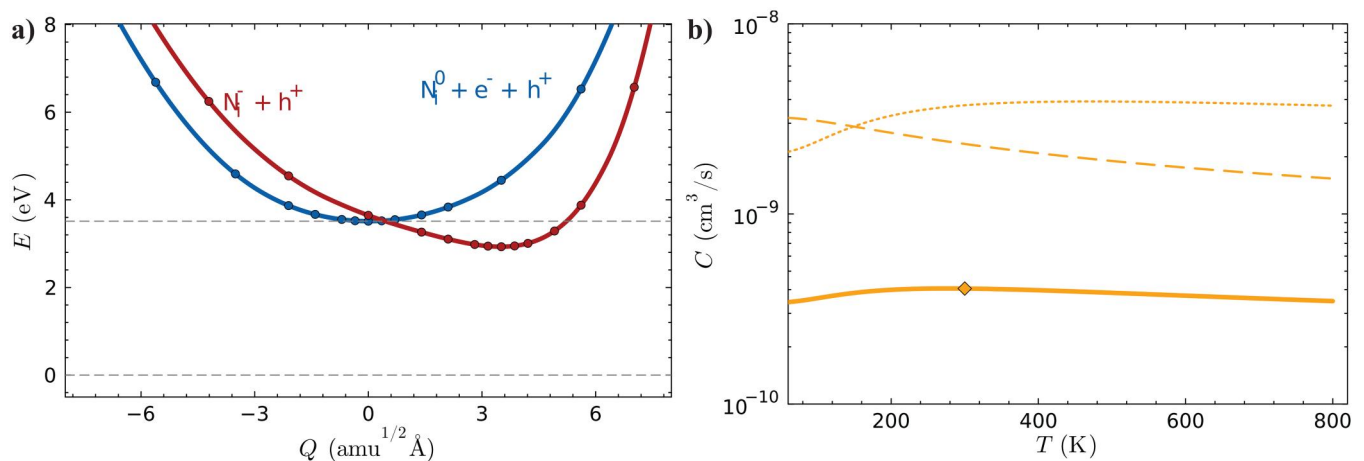


FIG. 14. (a) Configuration coordinate diagram for $N_{\text{Iz}} (0/-1)$. (b) Scaled electron capture coefficient for $N_{\text{Iz}} (0/-1)$ and sensitivity curves for $\Delta E \pm 0.05$ eV (dotted line is $+0.05$ eV and dashed line is -0.05 eV).

TABLE V. Nonradiative carrier capture timeframe estimate for V_{Ga} in p-type GaN, assuming $p = 10^{18} \text{ cm}^{-3}$.

q_1/q_2	$C_p \text{ (cm}^3/\text{s)}$	$t \text{ (s)}$
(-3/-2)	6.2×10^{-17}	1.6×10^{-2}
(-2/-1)	7.0×10^{-18}	1.4×10^{-1}
(-1/0)	1.4×10^{-14}	7.1×10^{-5}
(0/+1)	5.2×10^{-9}	1.9×10^{-10}

balance for a given system, using assumed or measured values for defect and carrier concentrations. The steady-state solution highlights which capture and emissions processes are dominant. In some cases, transitions with a large capture coefficient may not occur frequently if it must be preceded by another transition with a lower coefficient value (a rating limiting step). For example, the (+1/+2) transition for N_{Ga} has a $9.4 \times 10^{-14} \text{ cm}^3/\text{s}$ hole capture coefficient at room temperature. However, the defect is neutral for a large section of the middle of the bandgap and is likely to have to undergo the (0/+1) transition first. The hole capture coefficient for (0/+1) is over four orders of magnitude lower, $4.3 \times 10^{-18} \text{ cm}^3/\text{s}$, indicating that the N_{Ga} would remain neutral.

In general, vacancies and interstitial defects are expected to both be created at a high concentration⁶⁶ and experience carrier capture at higher rates. Under non-ionizing energy loss processes that create displacement damage, vacancies and interstitials are generated in Frenkel pairs as the primary defects, while complexes and antisites may be generated secondarily as the defects evolve. Because the bulk lattice is neutral, no extra charge is around, and no net charge transfer occurs during the first femtoseconds of the Frenkel pair formation. As it forms, the wavefunctions of the vacancy and interstitial overlap such that electrons are transferred from one to the other, maintaining overall charge neutrality. In order to match the Fermi level of the material, the defects then attempt to undergo transitions to reach their equilibrium charge state. However, the defects may remain in a non-equilibrium charge state for an extended period of time because of rate limiting transitions. Electrostatic attraction or repulsion between the point defects comprising the Frenkel pair may then enhance or suppress recombination.

During irradiation, Ga and N Frenkel pairs and other defect complexes can be generated. A full analysis of the evolution and annealing of defects in the post-damage cascade is beyond the scope of the present study. The timescale of free carrier recombination with point defects in GaN is estimated below. Another important issue will be charged defect diffusion and defect-defect reactions, such as a positively charged N_i diffusing to annihilate a negatively charged V_{Ga} , forming N_{Ga} . The formation of new point defects and complexes through diffusion like this changes the available transitions, reducing or increasing the likelihood for carrier capture to occur. The information provided here could be implemented in classical molecular dynamics simulations including charge state variations or combined with charged defect diffusion data in a dynamical multiscale atomistically informed device engineering method. The latter has been implemented by Diaz *et al.* to examine post-irradiation processes in gallium arsenide.⁸²

Considering an example for a p-type material, one can identify how long it would take a negatively (-3) charged V_{Ga} to become positively charged (+1) at 300 K. Assuming post-irradiation conditions with a region of many available electron-hole pairs, a charge carrier concentration of $n = p = 10^{18} \text{ cm}^{-3}$ was used²² and carrier mobility was not considered. It would take approximately 0.16 s to go through all four transitions. Table V shows the individual time estimations for each transition, indicating that the (-2/-1) step is the rate limiting step, taking 0.14 s. At the same time, the transitions between the singly charged states and the neutral state are the quickest and happen on the time scale of micro- to nano-seconds. In this scenario, the timescale for the V_{Ga} to transition past the -3 and -2 charge states is much greater than the expected recombination timescale. Therefore, with a positively charged Ga_{Li} , recombination of the Frenkel pair will be enhanced.

Due to their position near 2.3 eV, both (+3/+2) and (+2/+1) transitions of Ga_{Li} will capture electrons for essentially any level of n-type GaN during charge equilibration. In a highly n-type material, if the Fermi level is very close to the CBM (i.e., within 0.3 eV), significant electron capture will occur due to the Ga_{Li} (+1/0) and V_{N} (0/-1) transitions. The lower transitions in the bandgap, V_{N} (+1/0) and N_{Li} (0/-1) electron capture, will also occur. However, when the Fermi level is lower (i.e., within 0.6 eV of the CBM), these two will become more primary processes. In a p-type material, the Fermi level is such that V_{Ga} (0/+1) hole capture will contribute most out of those above $10^{-10} \text{ cm}^3/\text{s}$ at room temperature. If the defects are originally neutral, V_{Ga} (0/+1) hole capture, N_{Li} (0/-1) electron capture, and V_{N} (0/-1) electron capture will contribute more. Importantly, these three defects are also expected to have the largest defect concentrations from displacement damage.⁶⁶

IV. CONCLUSION

Nonradiative carrier capture via multiphonon emission was modeled for each native point defect in GaN and for the $V_{\text{Ga}}-V_{\text{N}}$ divacancy. Capture coefficients were calculated via electronic structure techniques using the C_{N} defect as a benchmark. Aside from the C_{N} transitions, seven transitions exhibited capture coefficients above $10^{-10} \text{ cm}^3/\text{s}$ at room temperature and are therefore dominated by nonradiative processes. Specifically, these transitions are electron capture for Ga_{Li} (+3/+2), Ga_{Li} (+2/+1), Ga_{Li} (+1/0), V_{N} (+1/0), N_{Li} (0/-1), and V_{N} (0/-1) in addition to hole capture for V_{Ga} (0/+1).

While these are the overall most likely capture processes, whether or not they impact device performance most significantly also requires the transition to be available. If the Fermi level is such that the transition is not attempted or a rate limiting process holds the defect in a prior charge state, the transition may not occur as frequently. Subsequently, it will not result in as significant of an impact on carrier density in the device.

Additional work is being done to incorporate these carrier capture coefficient results into technology computer-aided design (TCAD) models. These models will be used to understand the expected impact of these competing carrier capture processes on device performance. Observing this impact will enable predictions

of carrier removal and the resulting threshold voltage and peak transconductance shifts. Studying these practical implications will connect this characterization of capture coefficients to the overall device degradation due to displacement damage.

SUPPLEMENTARY MATERIAL

See the [supplementary material](#) for additional information supporting the present work. Further information on formation energy calculations is provided. A table of all capture coefficients at three temperatures, parameters for the calculations, effective barrier fits, and details on the charge transitions not included in the manuscript is included as well.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Alexander Sebastian Hauck: Conceptualization (supporting); Data curation (lead); Formal analysis (lead); Investigation (lead); Methodology (lead); Software (lead); Validation (lead); Visualization (lead); Writing – original draft (lead); Writing – review & editing (equal). **Miaomiao Jin:** Conceptualization (supporting); Formal analysis (equal); Funding acquisition (equal); Methodology (equal); Project administration (equal); Resources (equal); Supervision (equal); Validation (equal); Writing – review & editing (equal). **Blair Richard Tuttle:** Conceptualization (lead); Formal analysis (equal); Funding acquisition (equal); Methodology (equal); Project administration (equal); Supervision (equal); Validation (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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